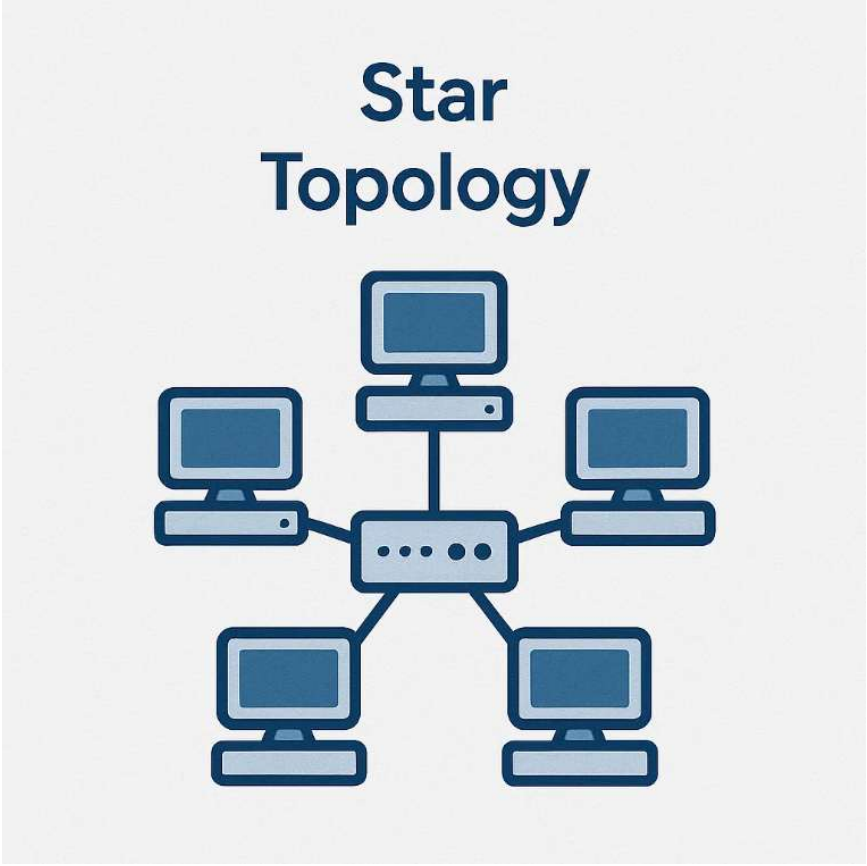
Assignment - 1.2

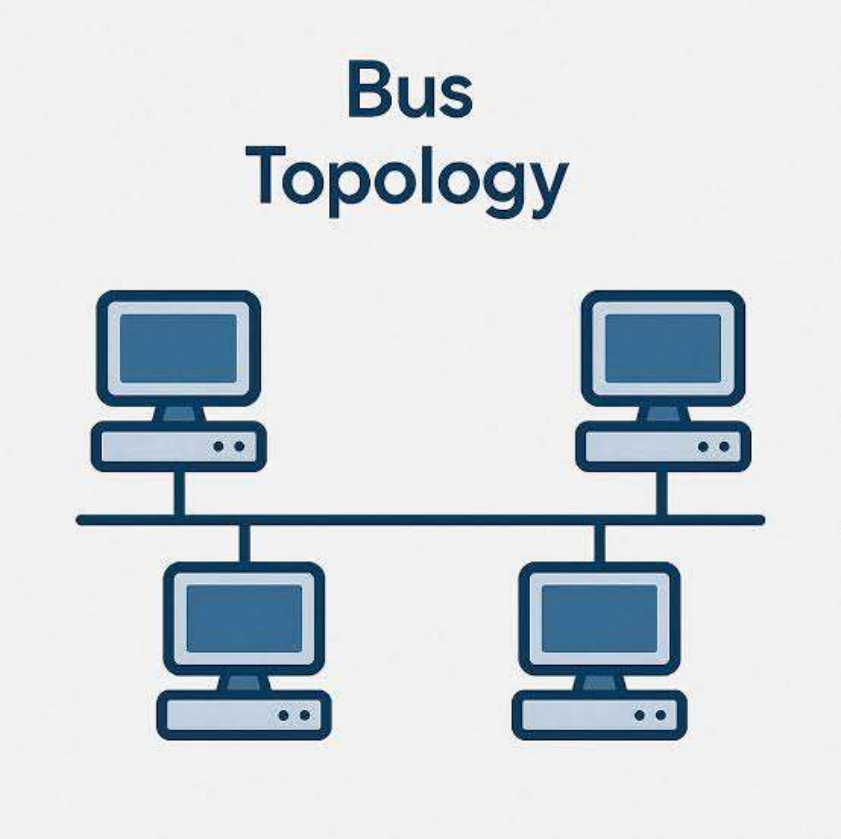
Network Topologies

Q - 1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.



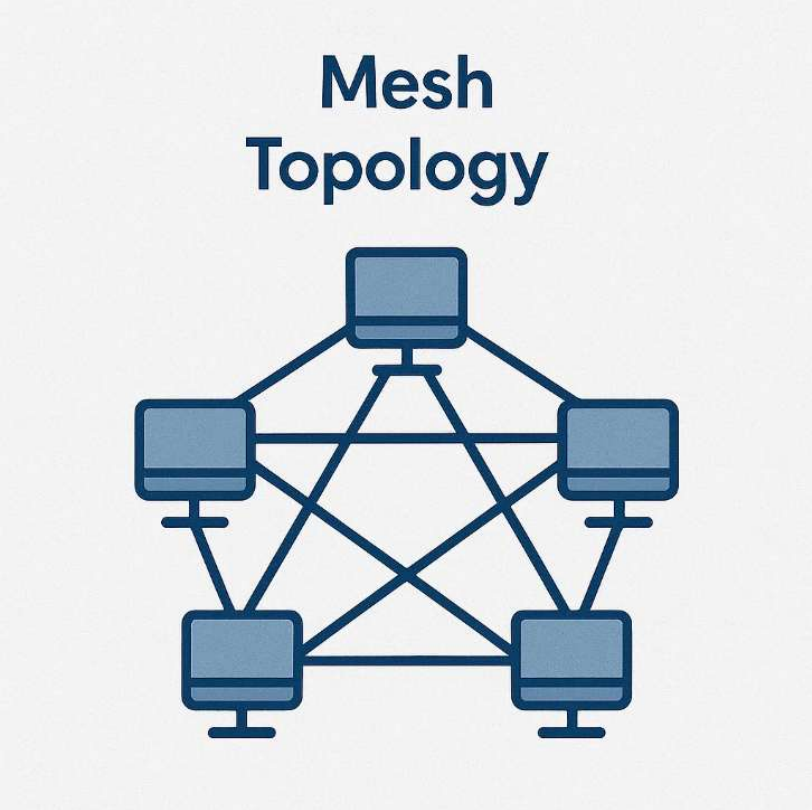
Application: Office computer network.

Explanation: All computers are connected to a central switch. If one computer fails, the others continue working.



Application: Small computer network.

Explanation: All computers are connected to one main cable. It uses less cable but failure of the main cable can stop the network.



Application: Internet network infrastructure.

Explanation: Devices have multiple connections. If one connection fails, data can travel through another available path.

Q - 2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case	App Example
Star	1. Easy to install. 2. Easy to manage. 3. Easy to find faults. 4. One device failure doesn't affect others.	1. Central device failure stops network. 2. More cables needed. 3. Costly. 4. Depends on central device.	Offices and schools	Instagram
Bus	1. Low cost. 2. Less cable needed. 3. Easy to set up. 4. No central device needed.	1. Main cable failure stops network. 2. Difficult to find faults. 3. Slow with many devices.	Small networks	Zomato

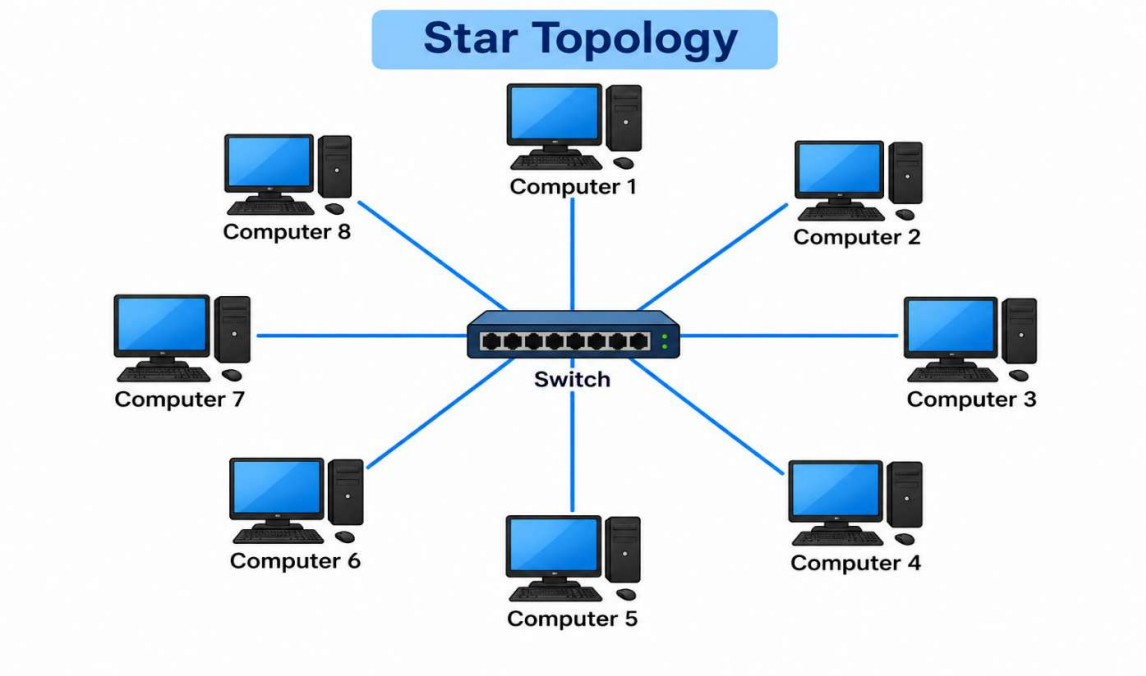
Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case	App Example
		4. Data collisions occur.		
Ring	1. Organized data flow. 2. Fewer collisions. 3. Equal chance to send data. 4. Predictable performance.	1. Link failure can stop network. 2. Difficult to add devices. 3. Hard to troubleshoot. 4. Data may take longer to travel.	Industrial networks	Paytm
Mesh	1. Highly reliable. 2. Multiple paths for data. 3. Link failure may not stop network. 4. Easy to find alternate paths.	1. Expensive. 2. More cables needed. 3. Difficult to manage. 4. Complex installation.	Large networks and internet backbone	WhatsApp

Q - 3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

- **Suitable Topology: Star Topology:**

1. **Speed:** A central game server helps players communicate quickly and reduces lag.
2. **Reliability:** If one player's connection fails, other players can continue playing.
3. **Cost:** It is cheaper and easier to set up than a full mesh topology.
4. **Example:** Games like PUBG and Free Fire use central servers to manage players, matches, and game data.

Q - 4. Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines. Hint: Consider factors like cost, ease of troubleshooting, and future expansion.



➢ **Explanation:**

Star topology is best because all 8 computers connect to a central switch. It is affordable, easy to troubleshoot, and simple to expand by connecting more computers. If one computer fails, the others continue working normally.